

Solution Requirements

Date	15 JULY 2026
Team ID	SWTID-2026-8038
Project Name	Credit card approval and Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)

1	Data Compliance and privacy	User Journey Phase/Stage: Stage 2 (Application Intake & Sensitive Data Upload) Design element description: End-to-end data encryption protocols, transparent user-consent popups, and automated masking of sensitive attributes (like SSN/National ID) during predictive feature processing	Low
2	Model Explainability	Integrating Explainable AI (XAI) frameworks (such as SHAP or LIME values) to generate real-time, plain-text reasons explaining exactly why an application was rejected or approved	High
3	Minimizing algorithms	<i>Pre-processing bias filters that remove proxy variables linked to protected classes (like gender, race, or zip code boundaries) from the prediction training set.</i>	High
4	Optimizing Predictive Accuracy	Utilizing balanced training datasets through advanced machine learning models (like XGBoost or Random Forest) optimized to reduce false rejections	Medium

5	Reporting	Automated API tracking modules that look beyond traditional bureau scores to evaluate alternative financial behaviors (like consistent rent or utility payment	high
---	-----------	--	------

Step 2: Non-Functional Requirements (NFR)

1	Data Privacy	Risk/Gain/Opportunity Category: Risk (Data Security) Design Element Description: Implement end-to-end data encryption, clear consent popups, and automated masking of sensitive attributes (like SSN) during feature processing.	Benefits/Priorities: Critical Priority. Ensures strict compliance with financial privacy regulations (like FCRA) and builds user trust during sensitive data collection.
2	Explainable Ai	quired "Adverse Action Notices" and provides clear reasons to rejected users.	Benefits/Priorities: Critical Priority. Ensures strict compliance with financial privacy regulations (like FCRA) and builds user trust during sensitive data collection.

3	Algorithm and fairness	Risk/Gain/Opportunity Category: Risk (Systemic Bias) Design Element Description: Pre-processing bias filters to scan and remove proxy variables linked to protected attributes (like race or gender) from the training data.	High Priority. Safeguards the institution against fairlending regulatory penalties and ensures fair evaluation for all demographic groups.
4	Error Optimization	Risk/Gain/Opportunity Category: Opportunity (Model Tuning) Design Element Description: Optimize predictive machine learning thresholds specifically to minimize false negatives (rejecting low-risk qualified applicants).	Benefits/Priorities: Medium-High Priority. Maximizes customer acquisition rates without increasing the bank's actual default risk profile.